Viruses

Viruses are:

- 1. Acellular
- 2. Obligate intracellular parasites
- 3. No ATP generating system
- 4. No Ribosomes or means of Protein Synthesis

Typical Virus   2 Parts

- 1. Nucleic Acid
  - DNA or RNA (But never both)
- 2. Capsid (Coat Protein)

- Some Viruses:
  - A. Envelope
  - B. Enzymes

Host range

- Spectrum of host cells that a virus can infect

- Some viruses only infect:
  - plants
  - invertebrates
  - protists
  - fungi
  - bacteria (Bacteriophages)

Host range

- Most viruses have a narrow host range
- Polio virus - nerve cells
- Adenovirus - cells in upper Respiratory Tract

Host range is determined by Viruses ability to interact with its host cell

- Binding Sites match Receptor Sites

- Binding Sites - on viral capsid or envelope
- Receptor Sites - on host cell membrane

TABLE 13.1 Viruses and Bacteria Compared

	Bacteria		Viruses
	Typical Bacteria	Rickettsias/ Chlamydias	
Intracellular parasite	No	Yes	Yes
Plasma membrane	Yes	Yes	No
Binary fission	Yes	Yes	No
Pass through bacteriological filters	No	No/Yes	Yes
Possess both DNA and RNA	Yes	Yes	No
ATP-generating metabolism	Yes	Yes/No	No
Ribosomes	Yes	Yes	No
Sensitive to antibiotics	Yes	Yes	No
Sensitive to interferon	No	No	Yes

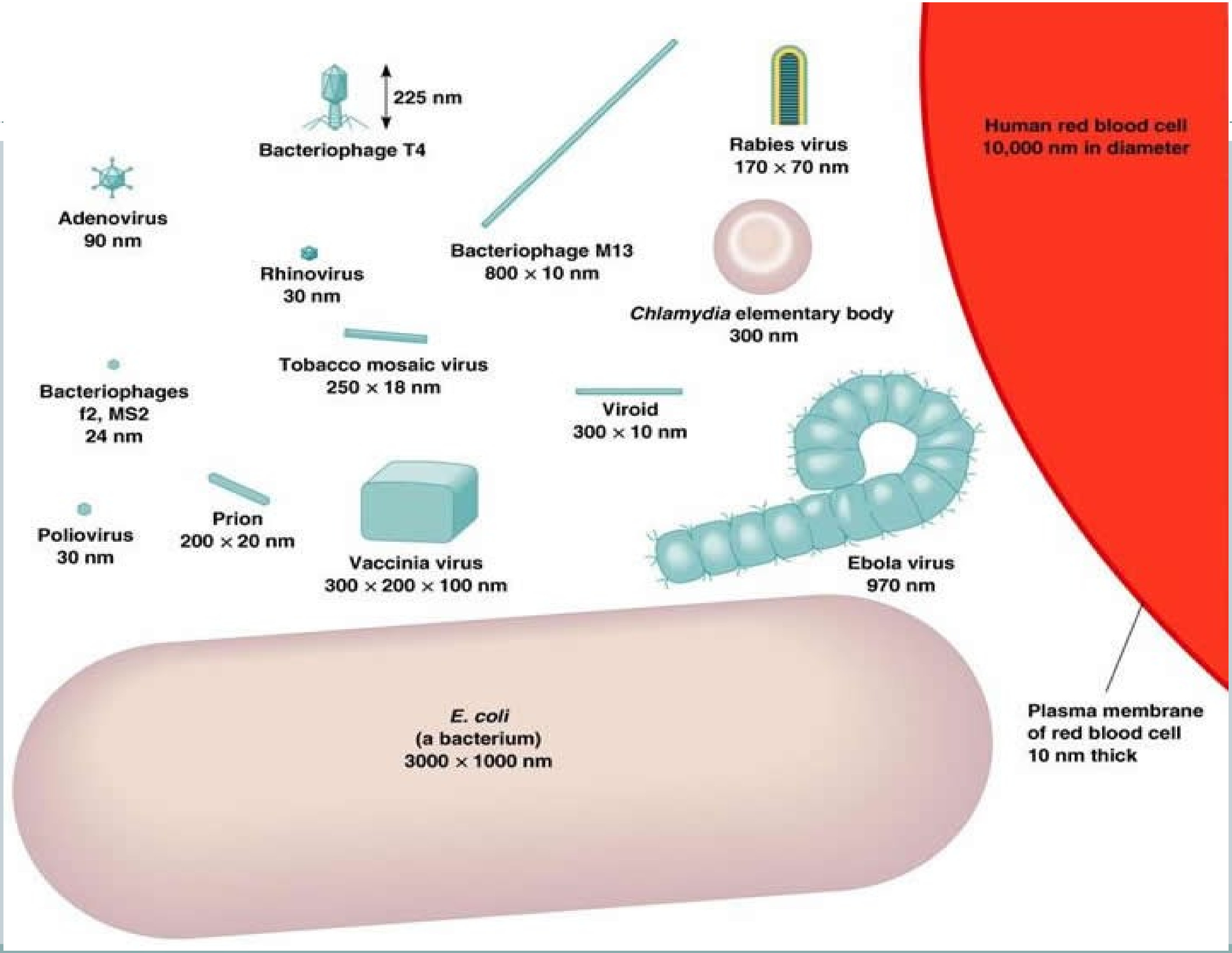
Viral Structure

- 1. Nucleic Acid
- 2. Capsid (Coat Protein)

- Nucleic Acid
  - DNA or RNA (But never both)
    - ssDNA
    - ds DNA
    - ss RNA
    - ds RNA

Viral Structure

- Capsid (Coat Protein)
  - protects viral genome from host endonucleases
  - capsomeres
  - Binding Sites
- Envelope
  - derived from the host cell
  - Binding Sites



225 nm
Bacteriophage T4
Rabies virus
170 × 70 nm
Human red blood cell
10,000 nm in diameter
Adenovirus
90 nm
Rhinovirus
30 nm
Bacteriophage M13
800 × 10 nm
Chlamydia elementary body
300 nm
Tobacco mosaic virus
250 × 18 nm
Bacteriophages
f2, MS2
24 nm
Viroid
300 × 10 nm
Poliovirus
30 nm
Prion
200 × 20 nm
Vaccinia virus
300 × 200 × 100 nm
Ebola virus
970 nm
E. coli
(a bacterium)
3000 × 1000 nm
Plasma membrane
of red blood cell
10 nm thick

- There is also a group of giant viruses, including the giant mimivirus, which is something like 800 nm in diameter.

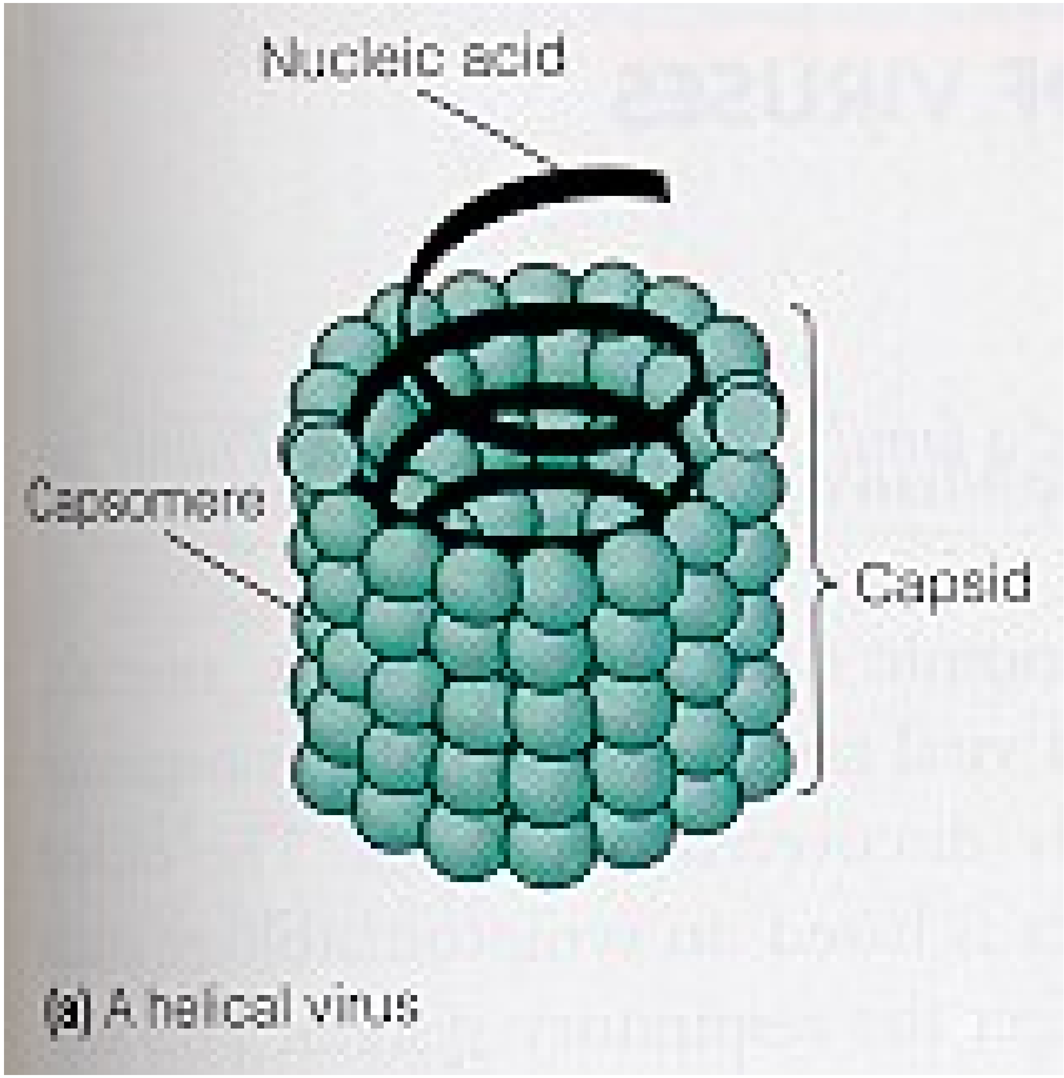
- ***Viral Structure***
- Virions are complete, fully developed viral particles composed of nucleic acid surrounded by a protein coat. Some viruses have an envelope composed of a phospholipid bilayer with viral glycoproteins.
- 1. Nucleic acid
- Viral genomes are either DNA or RNA (not both).
- Nucleic acid may be single- or double-stranded
- Nucleic acid may be circular or linear or

- Capsid - protein coat
- Capsomeres are subunits of the capsid
- Protomeres are capsomere subunits.

- . Envelope – the outer covering of some viruses, the envelope is derived from the host cell plasma membrane when the virus buds out. Some enveloped viruses have spikes, which are viral glycoproteins that project from the envelope.
- Influenzavirus has two kinds of spikes, H (hemagglutinin) and N (neuraminidase). The H spike allows the virus to attach to host cells (and red blood cells), the N spike is an enzyme that allows the mature viral particles to escape from the host cell
- Non enveloped or naked viruses are

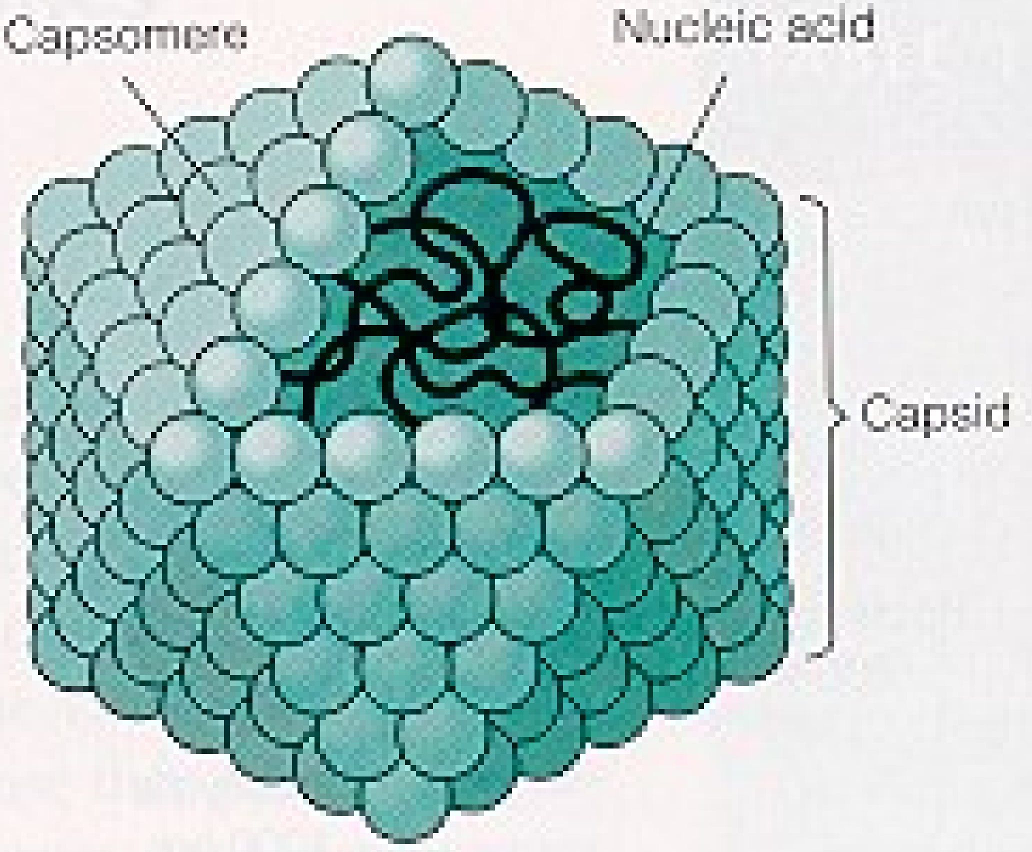
Viral Morphology

1. Helical



Viral Morphology

2. Polyhedral

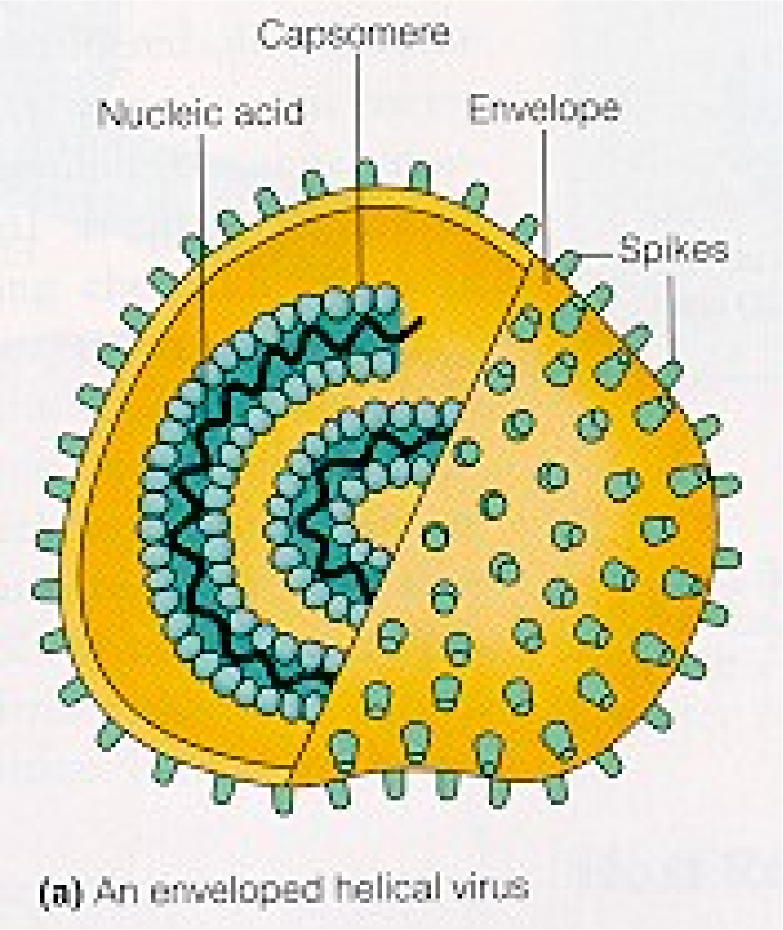


(a) A polyhedral virus

icosahedral

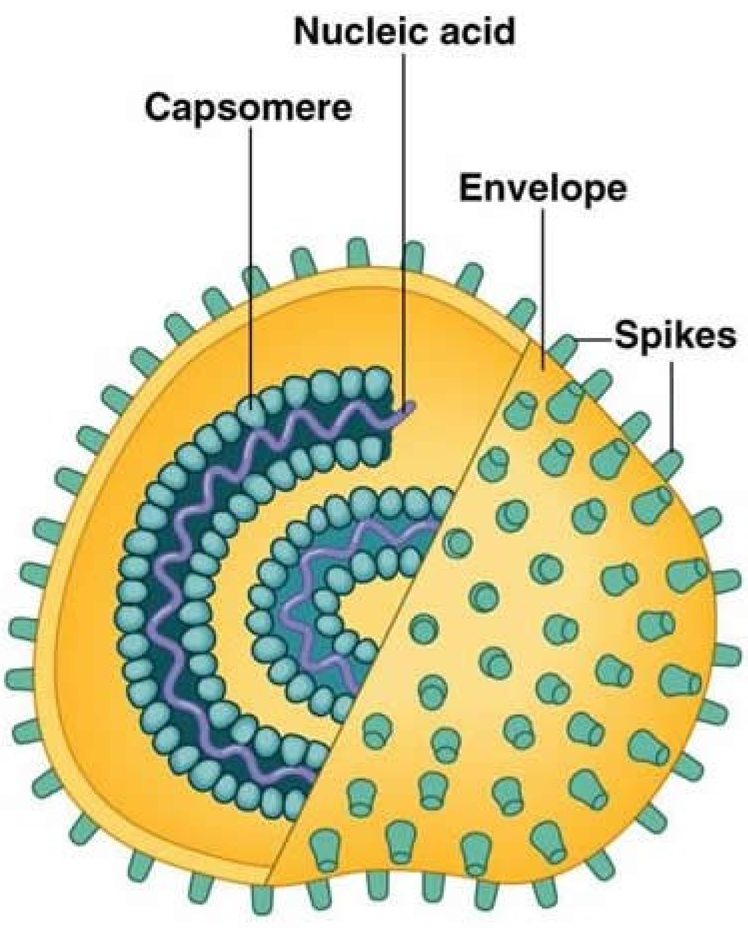
Viral Morphology

3. Enveloped

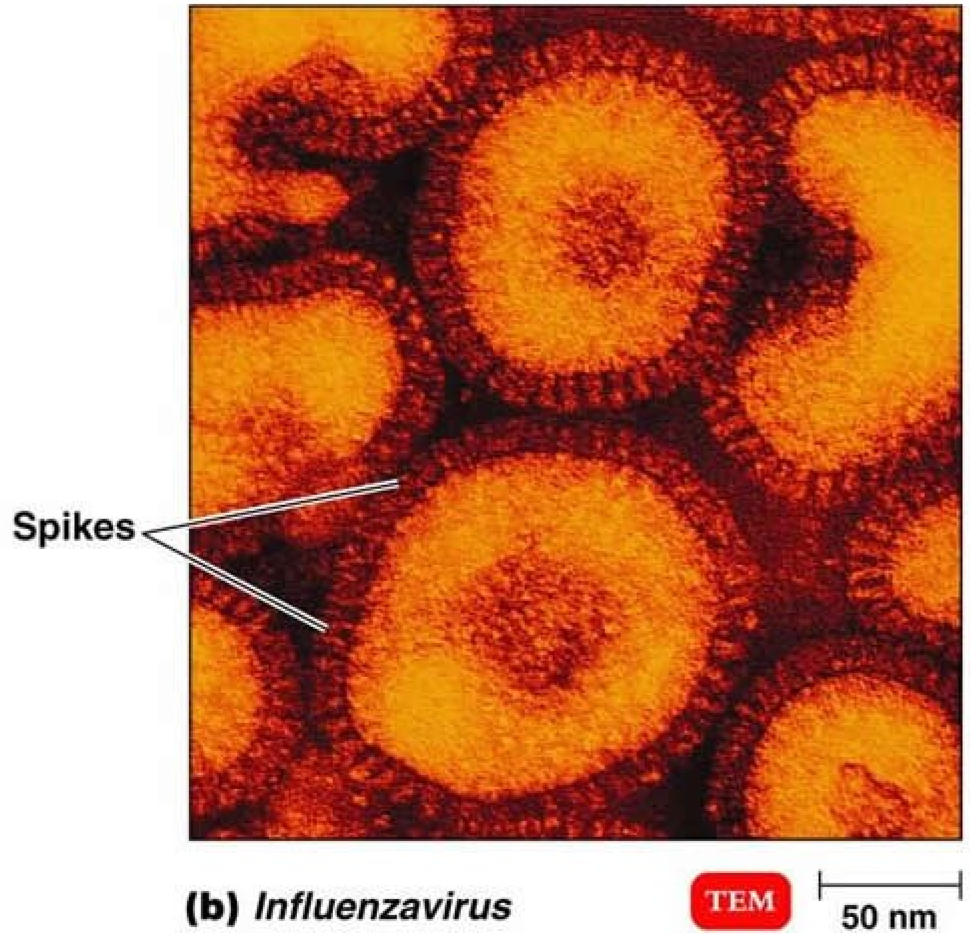


A. Enveloped Helical

B. Enveloped Polyhedral



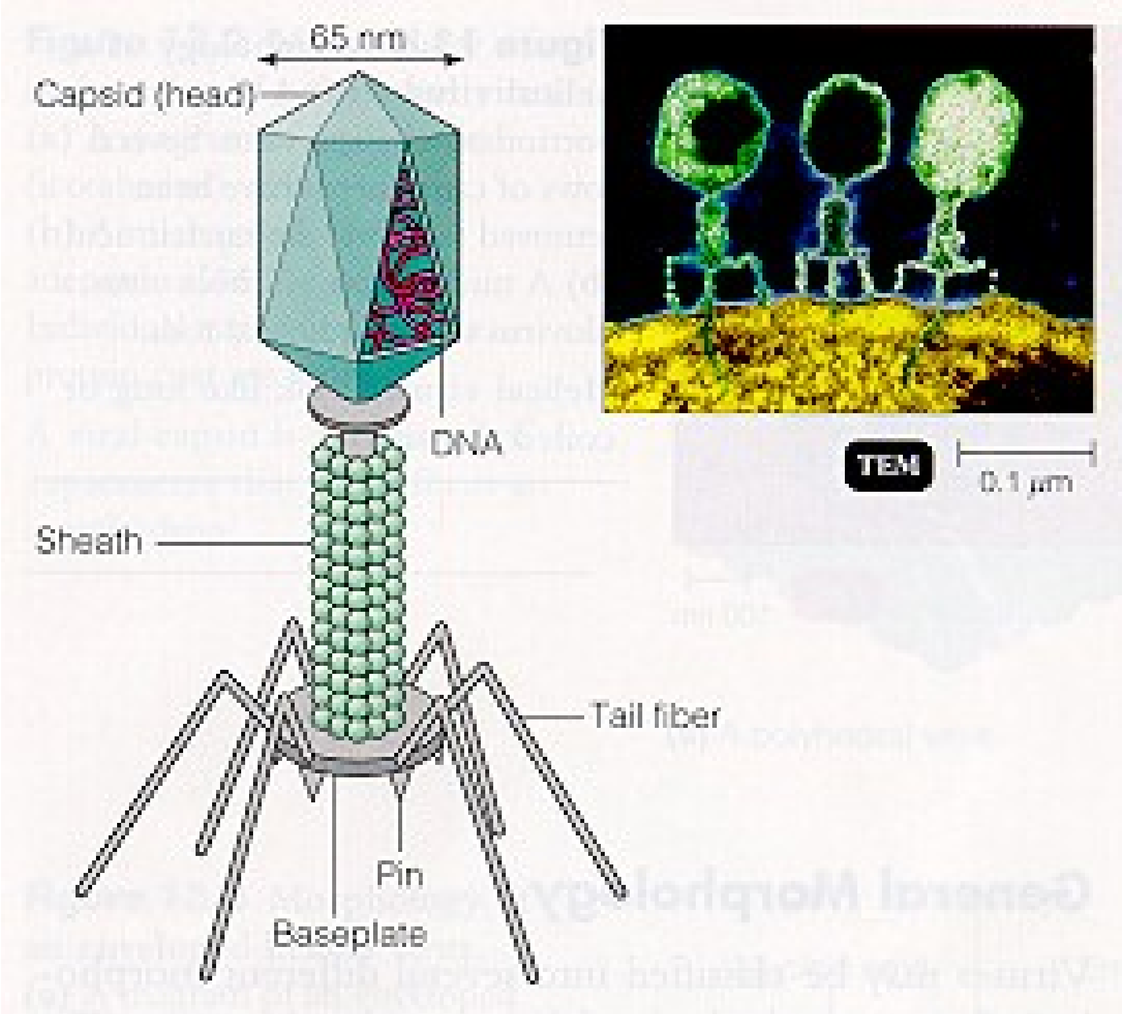
(a) An enveloped helical virus



(b) *Influenzavirus*

Viral Morphology

4. Complex



(a) A T-even bacteriophage

Viral Classification

- 1. Nucleic Acid
- 2. Morphology
- 3. Strategy for replication

Growing Viruses

- 1. Bacteriophages
  - Lawn of Bacteria on a Spread Plate
  - Add Bacteriophages
  - Infection will result in “Plaques”
    - Clear zones on plate

Growing Viruses

- Animal Viruses
  - A. Living Animals
    - mice, rabbits, guinea pigs
  - B. Chicken Embryos (Eggs)
    - used to be most common method to grow viruses
    - Still used to produce many vaccines (Flu Vaccine)
  - C. Cell Cultures
    - Most common method to grow viruses today

Viroids

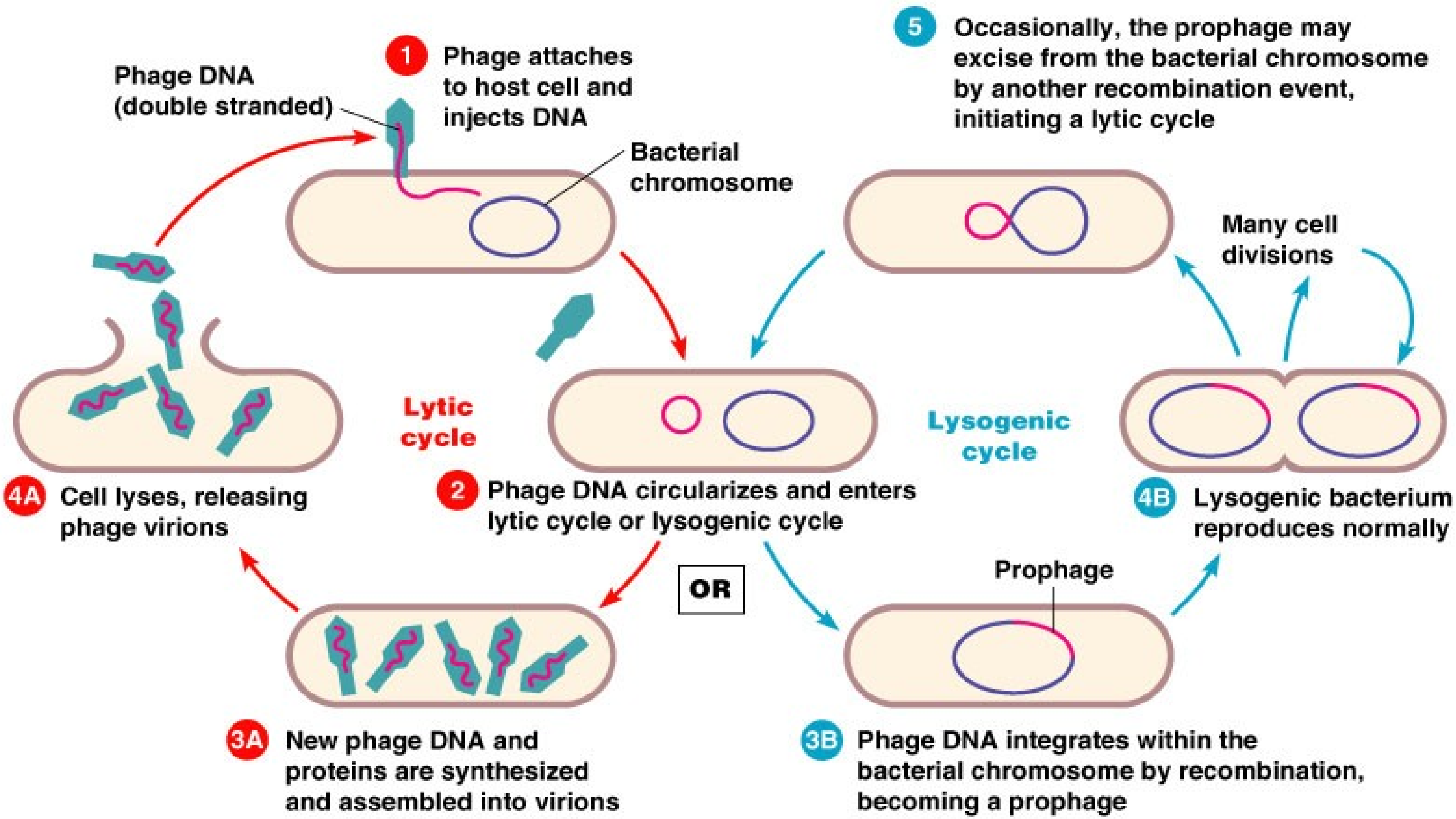
- Viroids
  - Naked RNA (no capsid)
  - 300 – 400 nucleotides long
  - Closed, folded, 3-dimensional shape (protect against endonucleases ?)
  - Plant pathogens
  - Base sequence similar to introns

Prions

- Proteinaceous infectious particle
- 1982
- Diseases
  - Scrapie (sheep)
  - Creutzfeldt-Jacob disease (CJD)
  - Kuru (Tribes in New Guinea)
  - Bovine Spongiform Encephalopathy (BSE)
    - Mad Cow Disease

Viral Replication

- Bacteriophage
  - 1. Lytic Cycle
  - 2. Lysogenic Cycle



1 Phage attaches to host cell and injects DNA
Phage DNA (double stranded)
Bacterial chromosome
5 Occasionally, the prophage may excise from the bacterial chromosome by another recombination event, initiating a lytic cycle
Many cell divisions
Lytic cycle
Lysogenic cycle
4A Cell lyses, releasing phage virions
2 Phage DNA circularizes and enters lytic cycle or lysogenic cycle
4B Lysogenic bacterium reproduces normally
OR
Prophage
3A New phage DNA and proteins are synthesized and assembled into virions
3B Phage DNA integrates within the bacterial chromosome by recombination, becoming a prophage

- **the Lyctic Cycle**
  The lytic cycle is thought to be the major method of viral replication as it results in total cell lyses, that is, cell destruction of the infected bacterium. The viruses that undergo lytic cycle are called virulent viruses. This is a six stage cycle that begins with the first stage called, 'penetration'. The virus injects it nuclei acids into the host cell that form a circle in the center of the cycle. The host cell is then tricked into replicating the viral nucleic acid instead of its own nucleic acids

- This viral DNA then begins organizing itself into a viral cell inside the host cell. When the number of viruses in the host cell increases, it causes the cell membrane to split under the pressure from so many cells. Thus, the viruses are released from the host body and ready to infect another cell. A host cell releases about 100 to 200 viruses approximately. These viruses cause cell lyses, thus giving rise to the name lytic cycle.

- Lysogenic cycle is the integration of the bacteriophage nucleic acid into the host genome. The new genetic material integrated into host cell is called a prophage. A prophage is transferred to the daughter cells of the host cell after each subsequent cell division.

- There are some viruses that replicate themselves by lysogenic cycle and carrying out partly lytic cycle. There are some DNA phages called the temperate phages that lyse small fraction of bacterial cells. The rest remain integrated in the bacterial genome. The viral nucleic acid is not expressed in the lysogenic state. The best example of a bacteriophage for understanding and striding lysogenic cycle is the lambda phage.

Lytic Cycle

- 1. Attachment- binding sites must match receptor sites on host cell
- 2. Penetration - viral DNA is injected into bacterial cell
- 3. Biosynthesis
  - Genome replication
  - Transcription
  - Translation

Virus uses Host Cells enzymes and machinery

Lytic Cycle

- 4. Assembly (Maturation)
  - viral particles are assembled
- 5. Release
  - Lysis

Lysogenic Cycle

- 1. Attachment
- 2. Penetration
- 3. Integration
  - Viral Genome is integrated into Host Cell Genome
  - Virus is “Latent”
  - Prophage

Lysogenic Cycle

- 4. Biosynthesis - Viral Genome is Turned On
  - Genome replication
  - Transcription
  - Translation
- 5. Assembly
- 6. Release
  - Lysis

Animal Virus Replication (non-enveloped virus)

- 1. Attachment
  - Binding Sites must match receptor sites on host cell
- 2. Penetration
  - Endocytosis (phagocytosis)
- 3. Uncoating
  - separation of the Viral Genome from the capsid

Animal Virus Replication (non-enveloped virus)

- 4. Biosynthesis
  - Genome Replication
  - Transcription
  - Translation
- 5. Assembly
  - Virus particles are assembled
- 6. Release
  - Lysis

Mumps, Measles, Influenza, and Poliomyelitis

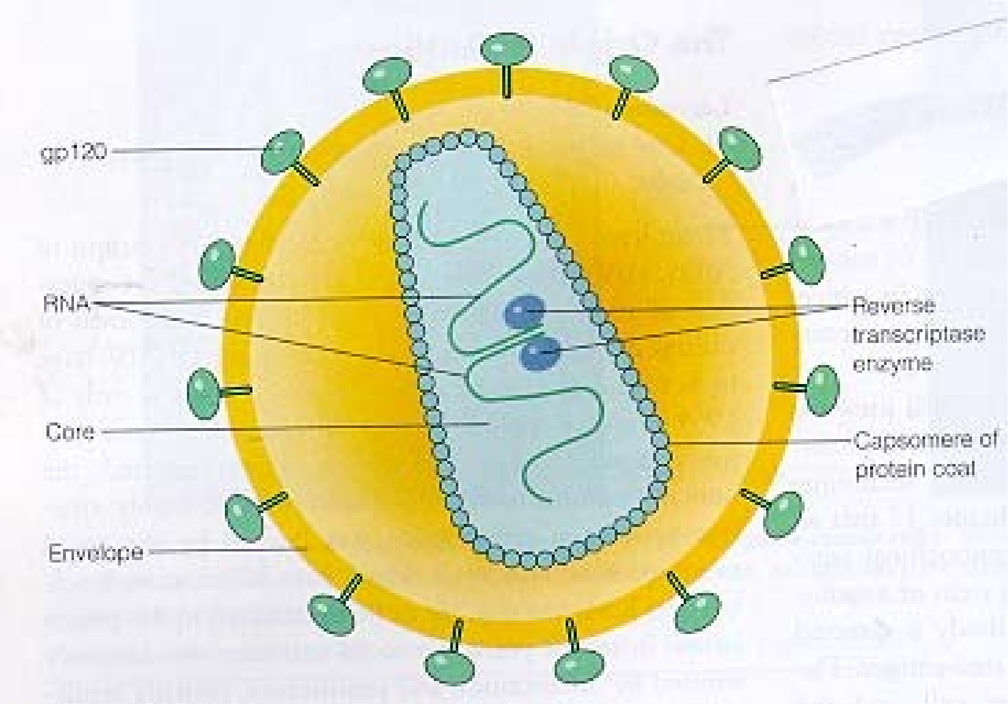
Retro Viruses

- 1. Many Cancer causing viruses
- 2. HIV
  - Human Immunodeficiency Virus
    - AIDS
      - Acquired Immunodeficiency Syndrome

HIV (Human Immunodeficiency Virus)

- AIDS
  - Acquired Immune Deficiency Syndrome
    - results in failure of the immune system
    - Death usually results from an Opportunistic Infection
- HIV discovered in 1984
  - By who ?
  - Luc Montagneir - Pasteur Institute

HIV Structure



(a) Structure of HIV

Retro Virus

Nucleic acid - RNA (2 strands)

envelope (120 binding sites)

Reverse Transcriptase

Modes of HIV Transmission

- HIV is transmitted by exposure to infected body fluids

- 4 Body Fluids
  - 1. Blood
  - 2. Semen
  - 3. Vaginal Secretions
  - 4. Breast Milk

How are these fluids transferred from one person to another?

- 1. High Risk Sexual Contact
  - unprotected vaginal sex
  - unprotected oral sex
  - unprotected anal sex
- 2. Needles
  - Intravenous Drug Abuse (sharing dirty needles)
  - accidental needle sticks

How are these fluids transferred from one person to another?

- 3. Blood to Blood Contact
  - open sores or wounds
  - Transfusions
  - Organ Transplants
  - Artificial Insemination
- 4. Mother to Child
  - placenta
  - as baby passes thru the birth canal
  - breast milk

Clinical Stages of an HIV Infection

- 1. Acute Infection
  - Initial infection of HIV (exposure to infected body fluids)
    - Fever
    - Headaches
    - Weakness
    - Muscle and joint aches
  - May last for a couple of weeks

2. Asymptomatic Disease

- Virus is “latent” inside CD4 cells
- Median latency period - 10 yrs.
- No signs or symptoms of illness (asymptomatic)
- HIV Positive - antibodies can be detected in your blood

3. Symptomatic Disease

- Viral Genome is "turned on", Symptoms begin to appear
- What causes HIV Genome to be turned on?
  - Other infections
  - stress
  - shock to the system
  - alcohol
  - drug abuse
  - nutrition
  - exercise ?

3. Symptomatic Disease

- Symptoms
  - chronic fatigue
  - low-grade fever
  - night sweats
  - diarrhea
  - weight loss
- Susceptible to Infections
  - bacterial pneumonia
  - meningitis
  - oral and vaginal yeast infections
  - tuberculosis

4. Advanced Disease (AIDS)

- Severe Opportunistic Infections
  - Pneumocysitis carinii pneumonia (PCP) Fungi
  - Kaposi's Sarcoma (Cancer - Skin and Blood vessels)
  - Toxoplasmosis (Brain) Protozoan
  - Cryptosporidiosis (G.I. Tract) Protozoan
  - Other Bacterial, Fungal and Viral Infections

Blood Test - ELISA

- Enzyme Linked Immunosorbant Assay
  - tests for HIV Antibodies
- If ELISA is positive, same sample is tested again
- If ELISA is positive again, then a Western Blot Test is done.
  - Western Blot - test for Viral antigens

Treatment for HIV Infection

- No Cure
- AZT (Azidothymidine)
  - Thymine analog
  - lacks a 3’ OH
  - Chain Terminator
  - Inhibits Reverse Transcriptase

AIDS Cocktail (Combination Therapy)

- AZT
- 3TC   (2’-deoxy-3’-thiacytidine)
- Protease Inhibitor

Vaccine for HIV ?

- HIV mutates too rapidly
  - Reverse Transcriptase causes at least 1 mutation each time it is used
    - 1 million variants during Asymptomatic Disease
    - 100 million variants during Advanced Disease (AIDS)